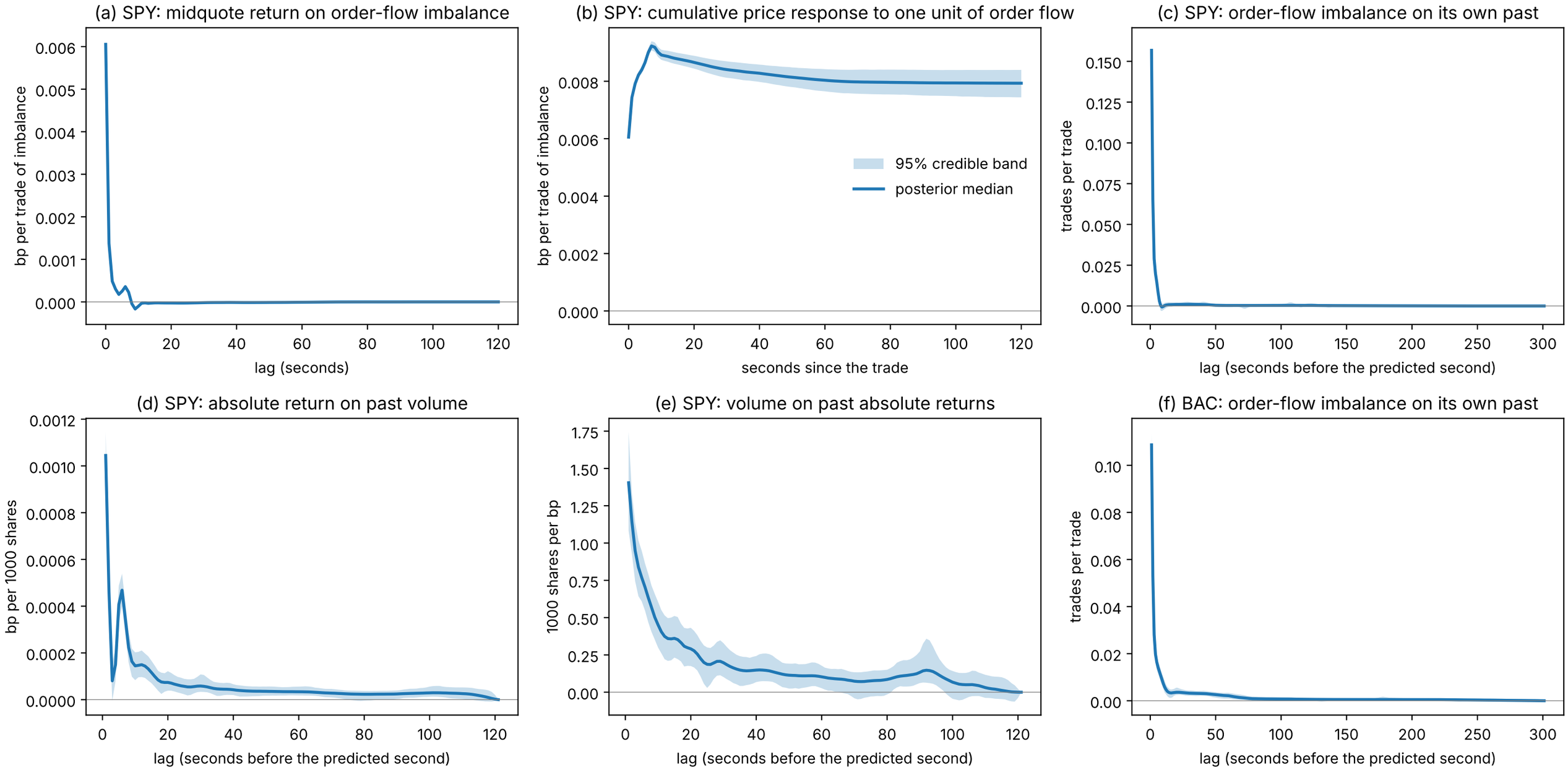
(a) SPY: midquote return on order-flow imbalance
bp per trade of imbalance
0.006
0.005
0.004
0.003
0.002
0.001
0.000
0
20
40
60
80
100
120
lag (seconds)
(b) SPY: cumulative price response to one unit of order flow
bp per trade of imbalance
0.008
0.006
0.004
0.002
0.000
95% credible band
posterior median
0
20
40
60
80
100
120
seconds since the trade
(c) SPY: order-flow imbalance on its own past
trades per trade
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
50
100
150
200
250
300
lag (seconds before the predicted second)
(d) SPY: absolute return on past volume
bp per 1000 shares
0.0012
0.0010
0.0008
0.0006
0.0004
0.0002
0.0000
0
20
40
60
80
100
120
lag (seconds before the predicted second)
(e) SPY: volume on past absolute returns
1000 shares per bp
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
20
40
60
80
100
120
lag (seconds before the predicted second)
(f) BAC: order-flow imbalance on its own past
trades per trade
0.10
0.08
0.06
0.04
0.02
0.00
0
50
100
150
200
250
300
lag (seconds before the predicted second)